

Evaluating Matrix Factorization Robustness in Highly Sparse Movie Recommendation Datasets

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Summary—This study evaluates three matrix factorization models—Funk Singular Value Decomposition (Funk SVD), Latent Factor Models (LFM-SGD), and Sparse Coding—for movie recommendation. Models were validated using RMSE and other regression-based metrics to assess prediction accuracy, alongside ranking metrics to evaluate recommendation quality. Experimental results indicate that Sparse Coding consistently outperforms the alternative techniques across all metrics, showing it is better at predicting ratings and works more reliably on new, unseen data.

I. INTRODUCTION

Matrix factorization is essential for mitigating missing data effects in recommendation systems. This study compares three models: **Funk Singular Value Decomposition (Funk SVD)**, **Latent Factor Models (LFM-SGD)**, and **Sparse Coding**.

The primary objective is to predict user ratings despite the extreme sparsity of interactions. We leverage dimensionality reduction to map users and items into a shared latent space, and handling the problem of missing information. Performance is evaluated via **RMSE**, **MAE**, **Precision**, **Recall**, and **NDCG**.

While **SVD** and **LFM-SGD** are conventional models, we hypothesize that **Sparse Coding** is better suited for sparse data. By using an overcomplete dictionary and regularization, it focuses on the most important data and ignores errors, capturing subtle preferences that traditional global-variance methods may overlook.

Facing a 93.7% sparsity rate, traditional models often falter. This study evaluates the resilience of **Sparse Coding** in extracting meaningful patterns from such a fragmented datasets.

A. Dataset Structure

This study utilizes the **MovieLens 100K** [1] dataset, which consists of 100,000 ratings ($R \in \{1, 2, 3, 4, 5\}$) provided by 943 users across 1,682 movies. The dataset exhibits a sparsity common in recommendation tasks, with a density of 6.304%.

The data is partitioned into training and testing subsets to facilitate model validation. Each observation is structured in a four-column format: 'user item rating timestamp'.

Auditing of the entries confirmed a clean feature space, allowing for direct mapping of user-item interactions into the latent factor models.

II. METHODS

A. Model Preparation

The models were trained on a sparse **User-Item interaction matrix** [1] of 943 users (U) and 1,682 movies (I). Using pre-partitioned subsets, the models learned latent factors to predict ratings that were kept hidden during training.

B. Performance Evaluation

To identify which model best captures subtle preferences in sparse data, performance was measured using several predictive and ranking metrics. Alongside **Mean Absolute Error (MAE)** [6], the primary error metric was **Root Mean Square Error (RMSE)** [1]:

$$RMSE = \sqrt{\frac{\sum_{u \in U, i \in I} (R_{ui} - R'_{ui})^2}{\text{Number of Ratings}}} \quad (1)$$

Where R_{ui} and R'_{ui} represent the actual and predicted ratings for user u and item i , respectively.

To assess ranking quality, we employ **Normalized Discounted Cumulative Gain (NDCG)** [7], which measures the effectiveness of the model in placing highly-rated items at the top of the recommendation list.

Unlike binary metrics, our implementation of NDCG utilizes the raw rating values as the relevance score rel_i , allowing the metric to account for the relative preference between different levels of observed user interest:

$$NDCG@k = \frac{1}{IDCG_k} \sum_{i=1}^k \frac{rel_i}{\log_2(i+1)}. \quad (2)$$

The score is normalized by the **Ideal DCG (IDCG)**, ensuring a result between $[0, 1]$. For this analysis, $k = 10$ is used as the cutoff rank, and the metric is only computed for users with ≥ 2 observed interactions to ensure a meaningful ranking evaluation.

To further assess relevance, **Precision** and **Recall** [6] were calculated using a threshold of 3, where ratings ≥ 3 are classified as “good” and others as irrelevant.

This multi-metric approach provides a robust evaluation framework by measuring point-prediction error, nuanced ranking order, and the binary accuracy of recommendations.

C. Matrix Factorization Models Used

- 1) **Funk Singular Value Decomposition (Funk SVD)** [3]: A latent factor method that optimizes solely over observed ratings, mapping users and items into a joint latent space via $\hat{r}_{ui} = q_i^T p_u$.
 - **Params:** $k = 50$, $\eta = 0.005$, $\lambda = 0.02$, 50 epochs.
 - **Uniqueness:** Focuses exclusively on pure user-item interactions without imputation or bias terms.
- 2) **Latent Factor Models (LFM-SGD)** [4]: An extension of the matrix factorization method that integrates baseline predictors to account for systematic biases: $\hat{r}_{ui} = \mu + b_u + b_i + q_i^T p_u$.
 - **Params:** $k = 20$, 20 SGD iterations.
 - **Uniqueness:** Disentangles general rating tendencies (global, user, and item effects) from specific user-item affinities.
- 3) **Sparse Coding** [5]: A generative method representing data as a sparse linear combination of atoms from an overcomplete dictionary. It employs Elastic Net regularization to isolate salient features from noise.
 - **Params:** $\lambda = 0.01$, $\alpha = 0.01$.
 - **Uniqueness:** Operates as a noise-filtering feature extraction method by enforcing structural sparsity rather than interaction-based factorization.

III. EXPERIMENTAL RESULTS AND METRIC DIVERGENCE

The experimental results as shown in Table I indicate a big difference between a model's ability to rank movies correctly and its ability to guess the exact rating.

While **Sparse Coding** achieves a near-optimal **NDCG of 0.99**, its **Norm_Accuracy** remains low at **0.13**.

TABLE I
MODEL PROFILE COMPARISON

Model	F1	Rec.	Prec.	N-Acc.	NDCG
Funk SVD	0.64	0.51	0.86	0.05	0.89
LFM-SGD	0.64	0.51	0.86	0.07	0.90
Sparse Coding	0.67	0.53	0.90	0.13	0.99

While **Sparse Coding** achieves a near-optimal **NDCG of 0.99**, its **Norm_Accuracy** remains low at **0.13**. This divergence suggests that the model effectively captures relative preference hierarchies (ranking) despite high absolute error in point-wise rating prediction ($Norm_Accuracy = 1 - RMSE_{test}$).

Similarly, **LFM-SGD** and **Funk SVD** demonstrate robust ranking capabilities with **NDCG** values of **0.90** and **0.89** respectively, yet yield negligible **Norm_Accuracy** scores (≤ 0.07).

This gap shows that while models understand which movies a user likes more than others, they struggle to hit adequate **RMSE**.

Consequently, these models are highly effective for recommendation retrieval where item ordering is prioritized over exact rating accuracy.

IV. DISCUSSION

A. Comparative Model Performance

The performance analysis in Figure 1 reveals a consistent trend in generalization, with test error remaining comparable to training error across all models.

This stability suggests that underlying patterns were captured without significant overfitting; however, **the persistent magnitude of the RMSE** indicates that the models have reached a predictive floor.

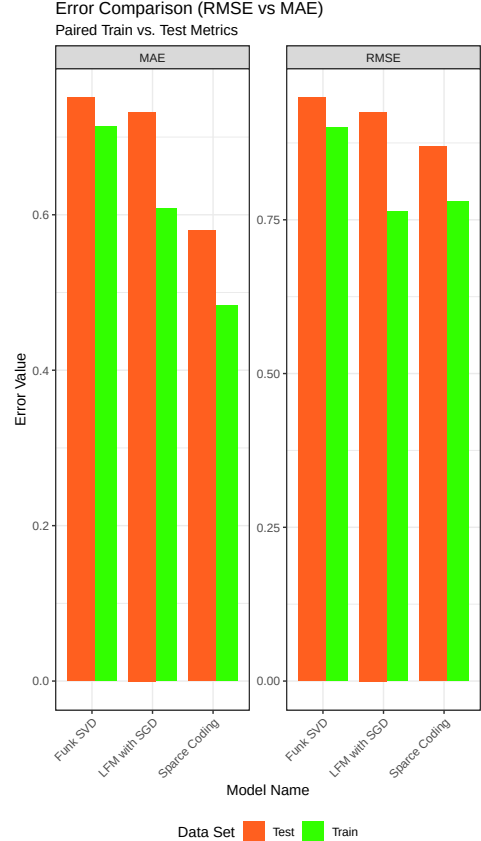


Fig. 1. Error Comparison by Model

Sparse Coding demonstrates the greatest predictive accuracy, achieving the lowest MAE and RMSE values on the test set.

Despite a higher training error relative to **LFM-SGD**, its superior performance on unseen data indicates a high capacity for generalizing user features to ratings without overfitting.

In contrast, **LFM-SGD** and **Funk SVD** exhibit higher baseline error rates. Although their training and test metrics are closely aligned—suggesting stability—their higher absolute values reflect lower precision in predicting exact ratings compared to **Sparse Coding**.

B. Model Generalization

Observing the slope graph in Figure 2 **Sparse Coding** achieved the highest performance, maintaining a stable NDCG of approximately 0.99 across both training and testing datasets. This minimal variance indicates exceptional generalization and high accuracy on unseen data.

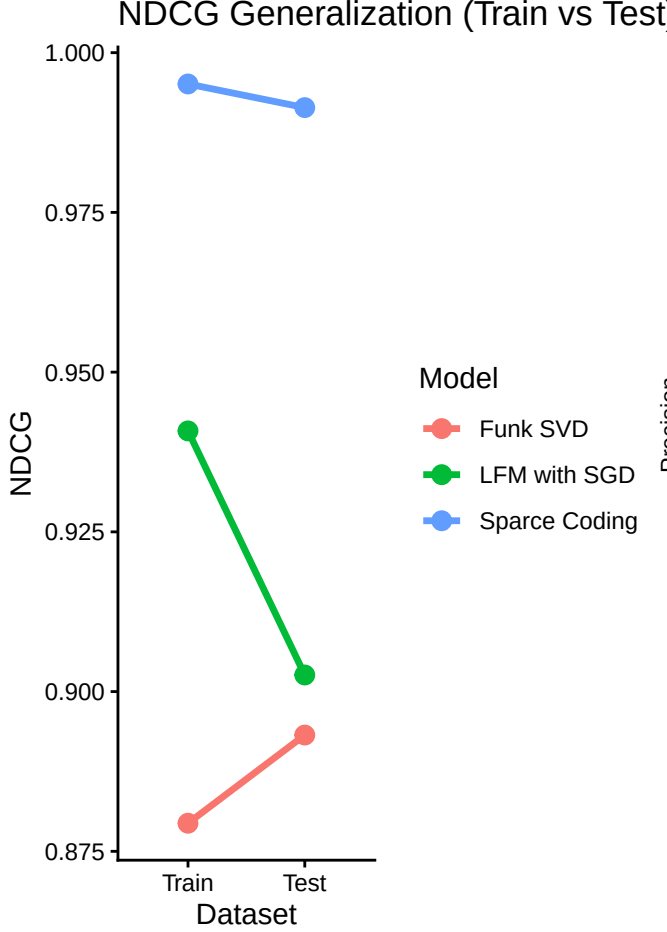


Fig. 2. NDCG by Model

Funk SVD exhibited a unique upward trend, with its NDCG rising from 0.88 during training to 0.89 in the testing phase. This slight improvement suggests the model is highly robust and effectively avoids overfitting.

LFM-SGD proved to be the weakest model in generalization. Its NDCG dropped sharply from 0.94 on the training set to around 0.90 on the test set.

This significant decline represents the largest generalization gap in the analysis, indicating substantial overfitting.

C. Evaluating Model Recommendation Efficacy

The visualization highlights shown in Figure 3 the trade-off between **Precision (the quality of recommendations)** and **Recall (the breadth of recommendations)**.

Sparse Coding emerges as the superior model, occupying the top-right position with a **Precision** ≈ 0.904 and **Recall** ≈ 0.534 , effectively maximizing relevance while minimizing noise.

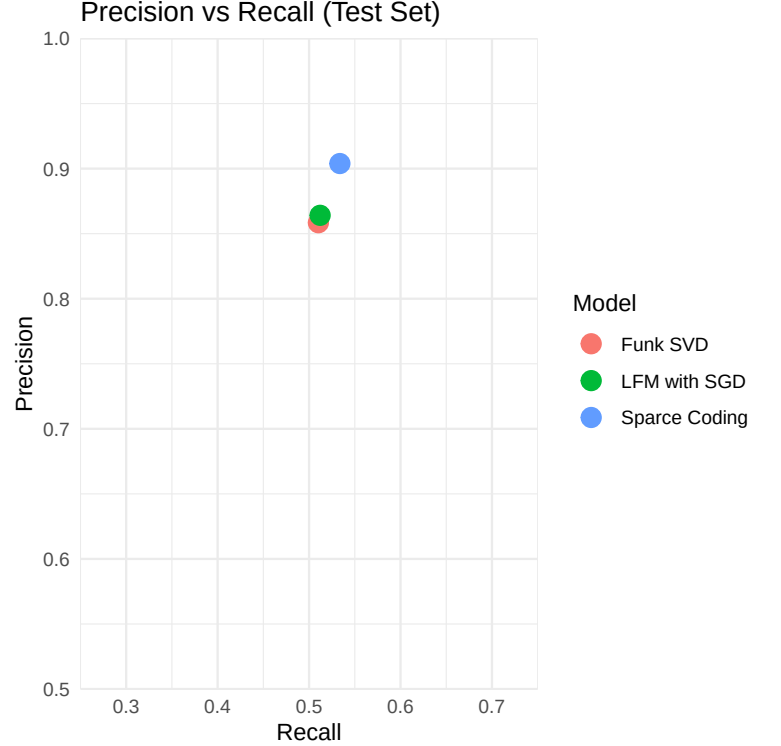


Fig. 3. Precision-Recall Comparison Among Models

LFM-SGD occupies the middle tier, yielding a **Precision** ≈ 0.864 and **Recall** ≈ 0.512 .

Finally, **Funk SVD** is the least performant model, sitting at the base of the cluster with a **Precision** of ≈ 0.858 and the lowest **Recall** at ≈ 0.51 , capturing the smallest portion of relevant items.

V. CONCLUSION

This study evaluated three matrix factorization models—**Funk SVD**, **LFM-SGD**, and **Sparse Coding**—on the MovieLens 100K dataset.

Despite the 93.7% sparsity and the complexity of the **latent factor space**, **Sparse Coding** emerged as the most robust framework, achieving superior precision and generalization.

A critical finding is the divergence between error-based and ranking-based metrics. While **RMSE** measures point-wise prediction magnitude, it proved insufficient as a sole indicator

of quality. Even as models reached a “predictive floor” in absolute accuracy, their ranking proficiency remained high.

The inclusion of **NDCG** was essential for this evaluation. While **RMSE** quantifies distance from actual values, **NDCG** rewards the correct ordering of recommendation lists.

Sparse Coding achieved a near-perfect **NDCG** of 0.99, demonstrating that it effectively find and recommend the right movies even when absolute predictions contain noise.

For real-world systems, where “top- N ” accuracy is paramount, ranking proficiency is a more vital indicator of user satisfaction than **RMSE**.

The superior performance of **Sparse Coding** across **Precision-Recall**, **NDCG**, and **RMSE**—relative to both **Funk SVD** and **LFM-SGD**—confirms its enhanced ability to separate the useful information from the random noise within sparse matrices.

Because **Sparse Coding** achieves high **NDCG** and **Precision**, it ensures that the ‘highest predicted ratings’ generated by the model truly correspond to the items the user is most likely to enjoy.

- **jsonlite**: Employed for the robust parsing and processing of JSON-formatted data.
- **dplyr** & **tidyr**: Used for data manipulation, cleaning, and reshaping into tidy formats.
- **ggplot2**: Used for the development of all statistical graphics and data visualizations.

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APPENDIX

APPENDIX A: SUPPLEMENTARY MATERIALS AND SOFTWARE USAGE

Source Code Repository

The source code, data, and detailed implementation scripts for this project are available on GitHub (URL: https://github.com/gaguillen4384-dev/Data-Science/tree/main/DataMining_2/Movie_Recommender_Model_Analysis).

Software Usage Acknowledgment

- 1) The text and documentation were edited and refined using **Google Gemini**.
- 2) **R** (version 4.x.x) was used for model computation and graphics development.
- 3) The following **R** libraries were essential to the computational workflow:
 - **recosystem**: Utilized for Latent Factor Modeling via Stochastic Gradient Descent (LFM-SGD).
 - **glmnet**: Applied for Sparse Coding through regularized generalized linear models.
 - **Matrix**: Used to manage and manipulate sparse matrix structures required for high-dimensional computations.